

Sebou Basin Experiment Setup Guide

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This document describes the scientific setup, datasets, and step-by-step workflow for running land surface model experiments (Open-Loop and three data assimilation variants) using Noah-MP/LIS over the Sebou River basin, Morocco.

1. Domain Configuration

Parameter	Value
Region	Upstream Sebou River basin + Saïss plain
South-West corner	33.0° N, 7.0° W
North-East corner	35.0° N, 4.0° W
Resolution	0.01° (~1 km)
Grid size	200 × 300 = ~60,000 active land cells
Simulation period	January 1, 2015 — December 31, 2020 (6 years, matching Nie et al., 2022)
Timestep	15 minutes

2. Scientific Objectives

The experiments address two key research questions for Q1 publications: 1. **Hydrology (Paper 1):** *Can joint soil moisture and LAI assimilation improve streamflow, AET estimation, and root-zone soil moisture representation?* 2. **Agriculture (Paper 2):** *Does assimilation reduce irrigation water allocation uncertainties and crop stress detection times?*

3. Experiment Design

Four experiments are defined, all on the same Sebou basin domain, to allow controlled comparison:

Label	Name	DA Instances	Config	Output
OPL	Open-Loop	None	<code>lis.config.opl_sebou</code>	<code>experiments/ OPL_sebou/</code>
DA-SM	SMAP SM assimilation	1: SMAP(NASA) soil moisture	<code>lis.config.da_sm_sebou</code>	<code>experiments/ DA_SM_sebou/</code>
DA-LAI	MODIS LAI assimilation	1: MCD15A2H LAI	<code>lis.config.da_lai_sebou</code>	<code>experiments/ DA_LAI_sebou/</code>
DA-Joint	Joint SM + LAI assimilation	2: SMAP SM + MCD15A2H LAI	<code>lis.config.da_joint</code>	<code>experiments/ DA_Joint_sebou/</code>

All DA experiments use **EnKF with 20 ensemble members** and GMAO-scheme perturbations on forcing (precipitation, radiation) and state variables (soil moisture layers, LAI).

4. Data Download Instructions

Three Python scripts in `scripts/download/` query NASA's Common Metadata Repository (CMR):

Dataset	Script	SLURM job
MERRA-2 forcing	<code>download_merra2.py</code>	<code>job_download_data.sh</code>
SMAP SPL3SMP v009	<code>download_smap.py</code>	<code>job_download_data.sh</code>
MODIS MOD15A2H v061 tile h17v05	<code>download_modis_lai.py</code>	<code>job_download_data.sh</code>

```
sbatch scripts/jobs/job_download_data.sh
```

All scripts are configured for the target period (2015-01-01 to 2020-12-31).

5. Parameter Processing (LDT)

```
sbatch scripts/jobs/job_1_ldt_sebou.sh
```

Status: Completed — generated `data/lis_input.d01_sebou.nc`.

Domain parameter maps (saved in `postproc/figures/`): - **Topography:** [sebou_topography.png](#) - **Land Cover (MODIS IGBP):** [sebou_landcover.png](#) - **Soil Texture (STATSGO):** [sebou_soil_texture.png](#)

6. MODIS LAI Preprocessing (Required for DA-LAI and DA-Joint)

The LIS "MCD15A2H LAI" plugin reads global NetCDF4 files (`86400 × 43200` grid, EPSG:4326, 500 m resolution), not raw MODIS HDF tile files. A preprocessing step is required:

```
sbatch scripts/jobs/job_preprocess_modis_lai.sh
```

This runs `scripts/fix/preprocess_modis_lai.py` which: - Warps each HDF tile from Sinusoidal to EPSG:4326 using GDAL - Stamps the tile into a global grid (filled with `255` outside the tile) - Saves compressed NetCDF4 files to `data/observations/MODIS_LAI/processed/YYYY/MCD15A2H.006_LAI_YYYYD0Y.nc4` - Renames variables from `Band1/Band2` to `Lai_500m/FparLai_QC` using `ncrename`

Status: Running — SLURM job 7053114 submitted.

7. Running the Experiments

Step 1 — Open-Loop (no assimilation)

```
sbatch scripts/jobs/job_2_lis_opl_sebou.sh
# Output → experiments/OPL_sebou/
```

Step 2a — DA-SM (SMAP Soil Moisture only)

```
sbatch scripts/jobs/job_3a_lis_da_sm_sebou.sh
# Output → experiments/DA_SM_sebou/
```

Step 2b — DA-LAI (MODIS LAI only, requires preprocessing)

```
# Wait for job_preprocess_modis_lai.sh to complete, then:
sbatch scripts/jobs/job_3b_lis_da_lai_sebou.sh
# Output → experiments/DA_LAI_sebou/
```

Step 2c — DA-Joint (SMAP SM + MODIS LAI, requires preprocessing)

```
# Wait for job_preprocess_modis_lai.sh to complete, then:
sbatch scripts/jobs/job_3c_lis_da_joint_sebou.sh
# Output → experiments/DA_Joint_sebou/
```

Tip: For the 3-day test run (2020-06-01 to 2020-06-04), the configs are already set. For the full 2015-2020 production run, update the **Ending year/month/day** fields in the config files.

8. Perturbation Attributes

File	Used in
<code>data/pert_package/forcing_attribs.txt</code>	All DA experiments
<code>data/pert_package/forcing_pertattribs.txt</code>	All DA experiments
<code>data/pert_package/noahmp_sm_attribs.txt</code>	DA-SM
<code>data/pert_package/noahmp_sm_pertattribs.txt</code>	DA-SM
<code>data/pert_package/smap_attribs.txt</code>	DA-SM, DA-Joint
<code>data/pert_package/smap_pertattribs.txt</code>	DA-SM, DA-Joint
<code>data/pert_package/noahmp_lai_attribs.txt</code>	DA-LAI, DA-Joint
<code>data/pert_package/noahmp_lai_pertattribs.txt</code>	DA-LAI, DA-Joint
<code>data/pert_package/modis_lai_attribs.txt</code>	DA-LAI, DA-Joint
<code>data/pert_package/modis_lai_pertattribs.txt</code>	DA-LAI, DA-Joint

9. Verification & Validation Strategy

Hydrology Focus (Paper 1)

- **River Discharge:** Route runoff via HYMAP; compare with ABHS gauge records.
- **Evapotranspiration (ET):** Compare against FAO WaPOR (250 m) or GLEAM v3.
- **Terrestrial Water Storage:** Validate against GRACE/GRACE-FO monthly anomalies.

Agriculture Focus (Paper 2)

- **Irrigation Volumes:** Compare with ABHS regional water withdrawal estimates.

- **Crop Productivity:** Correlate simulated GPP anomalies with provincial wheat/ barley yield statistics (Ministry of Agriculture of Morocco).

1. Domain Configuration

The domain is expanded to cover the entire upstream Sebou River basin feeding the reservoir, as well as adjacent agricultural perimeters in the Saïss plain: * **Bounding Box coordinates:** * South-West Corner: 33.0° N, 7.0° W * North-East Corner: 35.0° N, 4.0° W * **Resolution:** 0.01° (~1 km spatial grid) * **Grid Dimensions:** 200 × 300 grid points (~60,000 active cells) * **Simulation Period:** January 1, 2015 — December 31, 2020 (6-year study period matching Nie et al., 2022).

2. Scientific Objective

The simulation aims to address key research questions for Q1 publications in hydrology and agriculture: 1. **Hydrology (Paper 1):** *Can joint soil moisture and LAI assimilation improve streamflow prediction, AET estimation, and root-zone soil moisture representation?* 2. **Agriculture (Paper 2):** *Does assimilation reduce irrigation water allocation uncertainties and crop stress detection times under severe droughts?*

3. Data Download Instructions

We have provided three python download scripts in `scripts/download/` to query NASA's Common Metadata Repository (CMR) and download the datasets using your Earthdata credentials: 1. **Meteorological Forcing (MERRA-2):** `scripts/download/download_merra2.py` 2. **Soil Moisture Observations (SMAP):** `scripts/download/download_smap.py` 3. **Leaf Area Index (MODIS LAI):** `scripts/download/download_modis_lai.py`

Downloading using SLURM:

To submit the downloading process to the Toubkal cluster:

```
sbatch scripts/jobs/job_download_data.sh
```

Note: The download scripts have been configured for the target study period (2015-01-01 to 2020-12-31).

4. Parameter Processing (LDT)

We have configured `configs/ldt.config.sebou` with the expanded Sebou basin coordinates. To process the elevation (DEM), soil textures, and land cover types:

```
sbatch scripts/jobs/job_1_ldt_sebou.sh
```

Status: Completed

This step was successfully run on Toubkal and generated the central parameter file: * `data/lis_input.d01_sebou.nc`

Visualizations of the domain parameters are saved under `postproc/figures/` and can be viewed directly: - **Topography Map:** [sebou_topography.png](#) - **Land Cover Map (MODIS IGBP):** [sebou_landcover.png](#) - **Soil Texture Map (STATSGO):** [sebou_soil_texture.png](#)

5. Model and Assimilation Configurations

The model config file `configs/lis.config.da_joint` is configured for joint EnKF assimilation: * **Ensemble Size:** 20 members (`Number of ensembles per tile: 20`) * **Dynamic Vegetation:** Enabled (`Noah-MP.3.6 dynamic vegetation option: 2`), allowing LIS to simulate prognostic phenology and calculate dynamic LAI. * **Data Assimilation Instances:** 2 instances: * *Instance 1:* SMAP L3 Soil Moisture (`SMAP(NASA) soil moisture`) * *Instance 2:* MODIS LAI (`MODIS LAI` , tile `h17v05`) * **Perturbation Attributes:** Configured to perturb precipitation/radiation forcing fields and model state variables (soil layers and Leaf Area Index) using the GMAO scheme. Perturbation parameters are defined in: * `data/pert_package/modis_lai_attribs.txt` * `data/pert_package/modis_lai_pertattribs.txt` * `data/pert_package/noahmp_lai_attribs.txt` * `data/pert_package/noahmp_lai_pertattribs.txt`

6. Verification & Validation Strategy

To publish in Q1 journals, you will validate the outputs (`experiments/DA_joint/`) against independent datasets:

Hydrology Focus

- **River Discharge:** Route LIS runoff outputs using the built-in **HYMAP** routing module and compare simulated daily/monthly streamflow against gauge records from the **Sebou Basin Agency (ABHS)**.
- **Evapotranspiration (ET):** Compare simulated actual evapotranspiration against **FAO WaPOR** (250 m) or **GLEAM v3** products.
- **Terrestrial Water Storage:** Validate basin-scale water changes against monthly **GRACE/GRACE-FO** anomalies.

Agriculture Focus

- **Irrigation Volumes:** Enable the Noah-MP sprinkler irrigation module and compare simulated seasonal water applications against ABHS regional water withdrawal estimates.
- **Crop Productivity:** Correlate simulated GPP (Gross Primary Productivity) anomalies during the growing season with provincial wheat/barley crop yield statistics from the **Ministry of Agriculture of Morocco**.

Evaluation Tooling

A custom Python visualization module (`scripts/plot_da_comparison.py`) has been provided to automatically parse the LIS NetCDF outputs across the four experiments (OPL, DA SMAP, DA LAI, DA Joint). * **Time Series:** Computes and plots the daily basin-averaged response of key hydrological variables (SSM, RZSM, Evapotranspiration, Runoff, LAI). * **Spatial Maps:** Generates detailed `Δ DA Joint - OPL` difference maps to visually inspect the exact spatial impact of the assimilated satellite retrievals. * **Note on LAI:** LAI plotting requires the `LAI` parameter in `configs/MODEL_OUTPUT_LIST.TBL` to be enabled (set to `1`) prior to running the simulations.

7. Pre-Production Workflow (2015-2020)

Before running the final 5-year Data Assimilation experiments, the following sequence of scripts must be executed to prepare the forcings and initialize the model states correctly (Spin-up).

1. **Download Forcing:** Run `sbatch scripts/jobs/job_4a_download_merra2.sh` to download the remaining MERRA-2 data for 2015-2020.
2. **Spin-up:** Run `sbatch scripts/jobs/job_4c_lis_spinup.sh` to execute an Open-Loop simulation for the 5-year period. This equilibrates the deep soil moisture and groundwater states. The `lis.config.spinup_sebou` is configured to save a restart file (`LIS_RST_NOAHMP36...`) at the end of the run.
3. **CDF Matching (SMAP):** Run `sbatch scripts/jobs/job_4b_ldt_cdf.sh` . (*Note: You must first configure `ldt.config.cdf` to point to the output history of the Spin-up run. This computes the bias correction scaling factors for SMAP.*)
4. **Production DA Initialization:** Once the Spin-up is complete, modify your Data Assimilation configs (`lis.config.da_*`) as follows:
 - Change `Start mode: coldstart` to `Start mode: restart`
 - Set `Noah-MP.3.6 restart file:` to point to the file generated in step 2.